

DV ICN Upgrade to GEN6



OPERATING DOCUMENTATION

5670087
Revision 3.0

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Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ (BG)

Това упътване за работа е налично само на английски език.

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод.
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告 (ZH-CN)

本维修手册仅提供英文版本。

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。

警告 (ZH-HK)

本服務手冊僅提供英文版本。

- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
- 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。
- 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。

警告 (ZH-TW)

本維修手冊僅有英文版。

- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

UPOZORENJE (HR)

Ovaj servisni priručnik dostupan je na engleskom jeziku.

- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod.
- Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik.
- Zanimarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

**VÝSTRAHA
(CS)**

Tento provozní návod existuje pouze v anglickém jazyce.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

**ADVARSEL
(DA)**

Denne servicemanual findes kun på engelsk.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

**WAARSCHUWING
(NL)**

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

**WARNING
(EN)**

This service manual is available in English only.

- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

**HOIATUS
(ET)**

See teenindusjuhend on saadaval ainult inglise keeles.

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

**VAROITUS
(FI)**

Tämä huolto-ohje on saatavilla vain englanniksi.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteen käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

**ATTENTION
(FR)**

Ce manuel d'installation et de maintenance est disponible uniquement en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

**WARNUNG
(DE)**

Diese Serviceanleitung existiert nur in englischer Sprache.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendienst-technikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

**ΠΡΟΕΙΔΟΠΟΙΗΣΗ
(EL)**

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

**FIGYELMEZTETÉS
(HU)**

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

**AÐVÖRUN
(IS)**

Þessi þjónustuhandbók er aðeins fáanleg á ensku.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

**AVVERTENZA
(IT)**

Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

**警告
(JA)**

このサービスマニュアルには英語版しかありません。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

**경고
(KO)**

본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

**BRĪDINĀJUMS
(LV)**

Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā.

- Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu.
- Neveiciet aprikojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

**[SPĖJIMAS
(LT)**

Šis eksploatavimo vadovas yra tik anglų kalba.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

**ADVARSEL
(NO)**

Denne servicehåndboken finnes bare på engelsk.

- Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

**OSTRZEŻENIE
(PL)**

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

**ATENÇÃO
(PT-BR)**

Este manual de assistência técnica encontra-se disponível unicamente em inglês.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

**ATENÇÃO
(PT-PT)**

Este manual de assistência técnica só se encontra disponível em inglês.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

**ATENȚIE
(RO)**

Acest manual de service este disponibil doar în limba engleză.

- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

**ОСТОРОЖНО!
(RU)**

Данное руководство по техническому обслуживанию представлено только на английском языке.

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

**UPOZORENJE
(SR)**

Ovo servisno uputstvo je dostupno samo na engleskom jeziku.

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

**UPOZORNENIE
(SK)**

Tento návod na obsluhu je k dispozícii len v angličtine.

- Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

**ATENCION
(ES)**

Este manual de servicio sólo existe en inglés.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

**VARNING
(SV)**

Den här servicehandboken finns bara tillgänglig på engelska.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

**OPOZORILO
(SL)**

Ta servisni priročnik je na voljo samo v angleškem jeziku.

- Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod.
- Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli.
- Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.

**DİKKAT
(TR)**

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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Revision History

Revision	Date	Description
1.0	March 5, 2014	Initial Release
2.0	March 1, 2016	Updated to DV25.1_R01, GEN6, and Dell R-630. Updated material return procedures.
3.0	February 23, 2018	Chapter 1: Revised harvest information. Added MyPLCReader App instructions.

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Chapter 1 Overview

1 ICN Upgrade Installation

1.1 Upgrade Introduction

This publication documents the mechanical installation information for upgrading the existing SUN ICN (GEN2, 3, 4) module or existing Dell 32 channel ICN (GEN5) in the PGR cabinet with the new Dell ICN (GEN6) module. The Dell ICN (GEN6) is only compatible with DV25.1_R01 and later software.



WARNING

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.



CAUTION

Possible personal injury!

The Sun 4170 ICN weighs approximately 40 pounds.

Two people are required to lift it.

1.2 Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period. File a report with:

- Call 1-800-548-3366. Select "Install Support Services for FOA and MIS"
- Contact your local service coordinator for more information on this process.

1.3 Product Locator

During installation, the Install FE shall confirm that the Model and Serial Numbers are part of the asset. If missing, use either the gib web interface (http://gib.gehealthcare.com/gib/gib_entry.jsp) or the MyPLCReader App. This information will then be loaded directly to Siebel by the IB Verification Team.

Use the following procedure to install the MyPLCReader App.

1. From the GE App Icon, search for and install the MyPLCReader App on your iPhone or iPad.

Illustration 1-1: MyPLCReader App



2. Update the Configuration Page.
 - a. The first time the App opens, the Configuration Page will open.
 - b. Enter your SSO, Country and **if in US select the edit button to change the default email address** (by default, this app will send the email to the Europe IB Team Resource.EuropeanCentralAdminGIB@ge.com). The email to be used in US is usa.ib.verification@ge.com.
3. Start your first Scan.
 - a. Go to the Menu, make the appropriate selection for the job you are performing. **New Install, Upgrade System, or Update System.**
 - b. Enter the applicable Data, Install End Date, System ID & RFS#.
 - c. Click **Scan Bar Codes** and start scanning each barcode on the PL Cards.
 - d. Click the “Email” Button.
 - e. Email is sent to the Default email address, which was set up earlier.

1.4 Required Equipment and Tools



NOTICE

Follow all required safety and PPE procedures customary for your organization when working on this product.

Table 1-1: General Safety Requirements

Item	Description
1	Wear composite safety shoes for foot protection.
2	Wear gloves for hand protection.
3	Wear safety glasses for eye protection.



FERROUS MATERIAL HAZARD!

IF MAGNET IS ENERGIZED, TOOLS, COMPONENTS, AND PARTS REQUIRED FOR THIS INSTALLATION THAT CONTAIN FERROUS MATERIAL WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

Tools to upgrade the system are listed in following table:

Table 1-2: Installation Equipment

Item	GE Part Number	Description
1		Standard Hand Tools
2		Screwdriver, slotted (long & narrow)
3	46-271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
4		Service Methods DVD
5	46-198094P1	Wrist grounding strap
6	---	Plastic or aluminum flashlight

1.5 Discarded Material Return Policy

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1.5.1 Material Policy Overview

1.5.1.1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly to Approved GE Recycling Center. Do not remove or swap any parts from the material to be returned.

1.5.1.2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items per local GEHC policy and procedures and other applicable published guidance. Contact GEHC Recycling Center for assistance.

1.5.2 Harvest Testing



NOTICE

The equipment must be tested prior to the system shutdown/upgrade to allow appropriate Harvest capabilities.

Before going on site:

- Print the proper regional address labeling from DOC2100599 Harvest Return Addresses.
- Verify that you can download the Harvest test procedure from MyWorkshop using DOC1899393.
- If you have not been contacted to test the system or cannot download the document, reach out to the local Harvest team. For all other requests contact:

@HEALTH Global Harvest Team GlobalHarvestTeam@ge.com

Once testing is completed, submit the test record for review. Once these test records are reviewed, they will be processed for electronic signatures.

Package the equipment and follow your regions method of returning harvest equipment.

1.5.3 Harvested Materials

Table 1-3: List of Materials to Be Harvested

System Type for Upgrade	MR450, MR450W, MR750 and MR750W- Test Procedure DOC1899393		
Subassembly	Part Number	Description	Qty
Parts to be returned are located in PGR Cabinet	5127452-2	Image Compute Node - Gen 2 with VRF	Up to 2
	5127452-6	Image Compute Node - Gen 2 with VRF and without Infiniband	1
	5337894	Image Compute Node - DV 16CH GEN 3	1
	5337894-2	Image Compute Node - DV 32CH GEN 3	1
	5911000	Image Compute Node - Gen 4.1 DV 16 Channel	1
	5911000-2	Image Compute Node - Gen 4.1 DV 32 Channel 32GB	1
	5911000-3	Image Compute Node - Gen 4.1 DV 32 Channel 72GB	1
	5921000	Gen5 16CH ICN	1
	5921000-2	DV Gen5 32CH ICN	1
	5921000-3	DV Gen5 32CH+ ICN	1
		Mounting brackets	Up to 4
		Cables	

NOTE: Reuse the packaging that the new Dell ICN was shipped in to return the removed Sun or Dell ICNs.

1.5.4 Return Kit

If needed, packaging and/or transportation details can be obtained through the local Installation Specialist.

Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.

1.5.5 Deinstallation Procedures

Refer to instructions in this manual for deinstallation procedures of site.

1.5.6 Product Locator Information

Process product locator information for all serialized components removed from the system according to policy. Refer to [Section 1.3](#).

1.6 Upgrade Components

The following components are provided for the Sun to Dell ICN upgrade:

- Dell ICN
- ICN Mounting Kit that includes:
 - Slide Bracket (2)
 - Slide Rail (1)
 - Nuts (8)
 - M5 x 12 Screw (8)
- Cable Kit that includes:
 - Ethernet Cables (3)
 - ICN Power Cord (1)

For upgrading an existing Dell 32 Channel to a Dell 32 Channel +, only the ICN will be upgraded. The provided slides, slide brackets, and cables are to be discarded.

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Chapter 2 Power Off

1 Power Off Procedure

1.1 Overview

This procedure documents the proper methods of shutdown for GEN2 to GEN4 ICNs. **NOTE:** The ICN modules are automatically powered on when the host computer at the GOC is powered on. There is no need to switch the ICNs on individually.

1.2 Safety



WARNING

ELECTROCUTION HAZARD!
DANGEROUS VOLTAGES ARE PRESENT THROUGHOUT THE SYSTEM CABINET WHEN POWER IS APPLIED.
REFER TO THE MR SERVICE SAFETY MANUAL, PN 5452735.



NOTICE

Follow this procedure for removing power to the ICN chassis. Failure to properly power down ICNs may result in premature ICN failure.



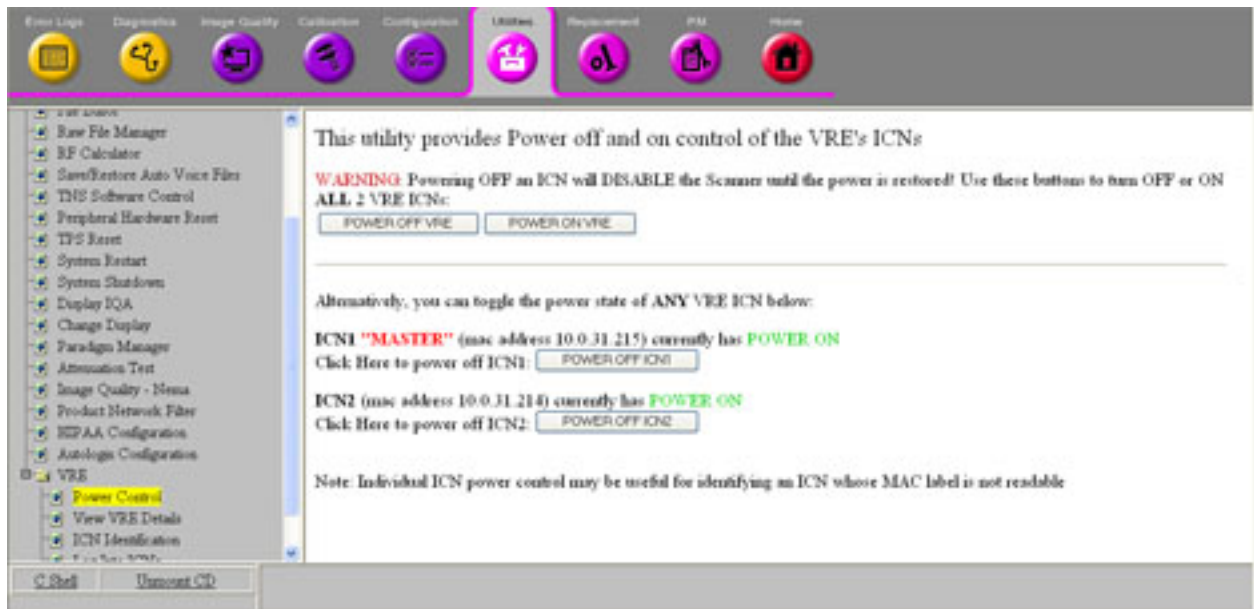
NOTICE

Do not turn power off to the ICNs using the power button on the front panel. The ICNs have software running on disk drives that may become corrupted if not powered down properly.

1.3 Power Down ICNs from VRE Power Control

1. At the host computer, on to the Service Browser, select *Utilities > Toolbox > VRE > Power Control*.

Illustration 2-1: Power Control



NOTE: The number configuration should match the number of VREs.

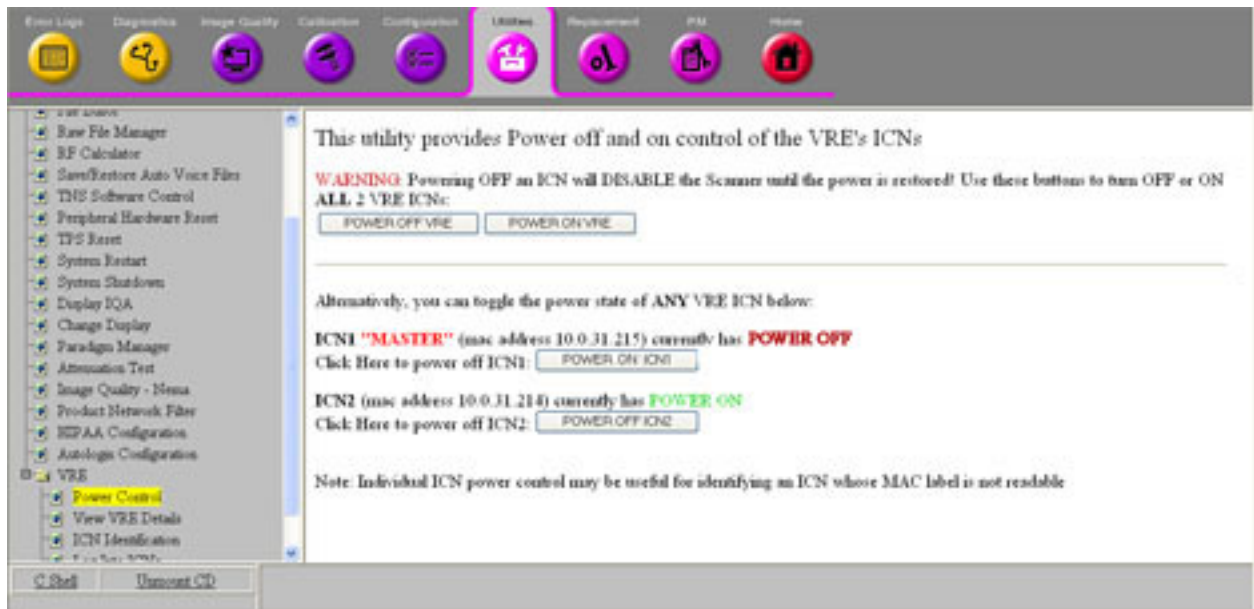
- Click [POWER OFF ICN(x)] (shown in previous illustration). Wait for the ICN to power off.

Illustration 2-2: ICN Switching Off Indication



When the Power Control link is refreshed, the screen shows the ICNs in their current state.

Illustration 2-3: ICN 1 Switched Off



NOTE: The illustration above shows images from a system with a Sun 4100 ICN. Systems with a Sun 4170 or 4170M2 ICN have only one ICN selection.

3. The Sun ICN(s) are powered down if the green disk drive LEDs on the front of the ICN(s) are no longer illuminated. The Power LEDs continue to blink even after power-down.

Illustration 2-4: ICN Front Panel (Sun 4100)

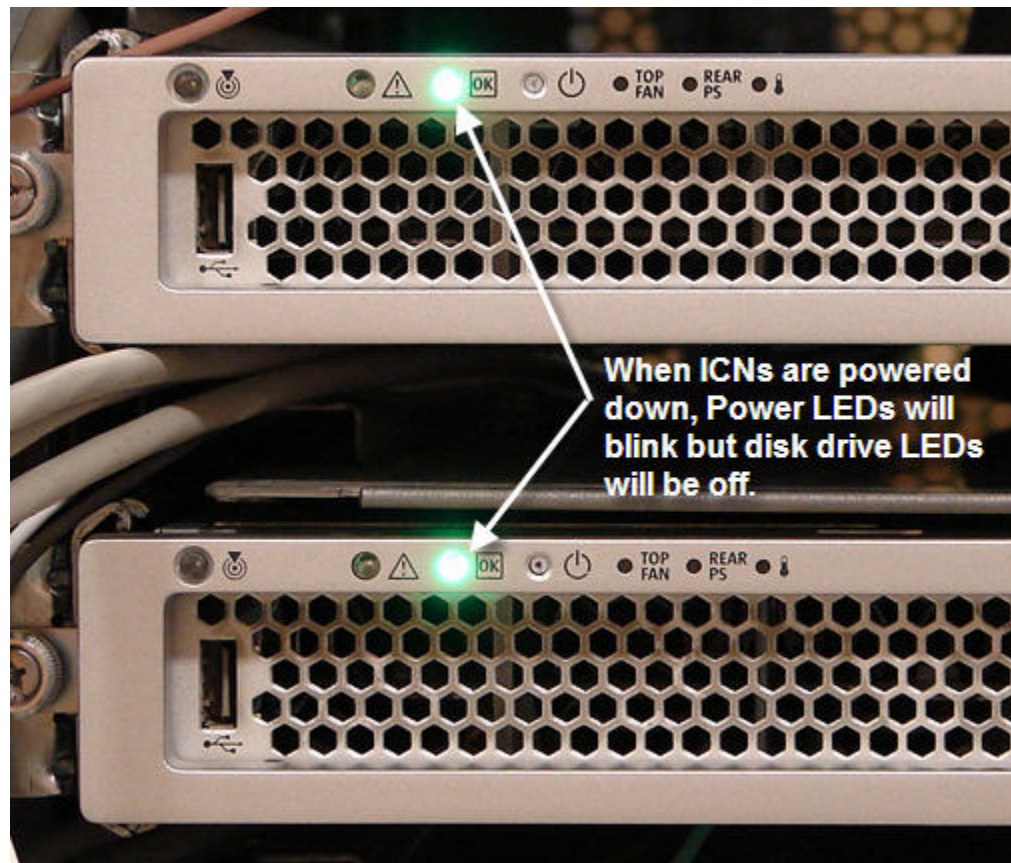


Illustration 2-5: ICN Front Panel (Sun 4170 and 4170M2)

When ICN is powered-down, Power LED will blink, but disk drive LEDs will be OFF.



4. The Dell ICN is powered down if the green power LED located on the upper left side of the front of the ICN is not illuminated.

Illustration 2-6: ICN front Panel for Dell R-630



1.4 Power Off ICNs Manually



NOTICE

If the browser or the command line are not accessible, the ICNs must be powered off manually. Try this method before powering off the GOC. This is the preferred method because it does not corrupt the files on the hard drive.

1. For Sun ICNs, momentarily press the power button. This brings down the ICN in a soft method. It takes about a minute or two for the operating system (OS) to boot down. (After power-down, the green power LED continues to slowly blink, but the disk drive LEDs go dark.)

Illustration 2-7: ICN Power LED and Power Button (Sun 4100)

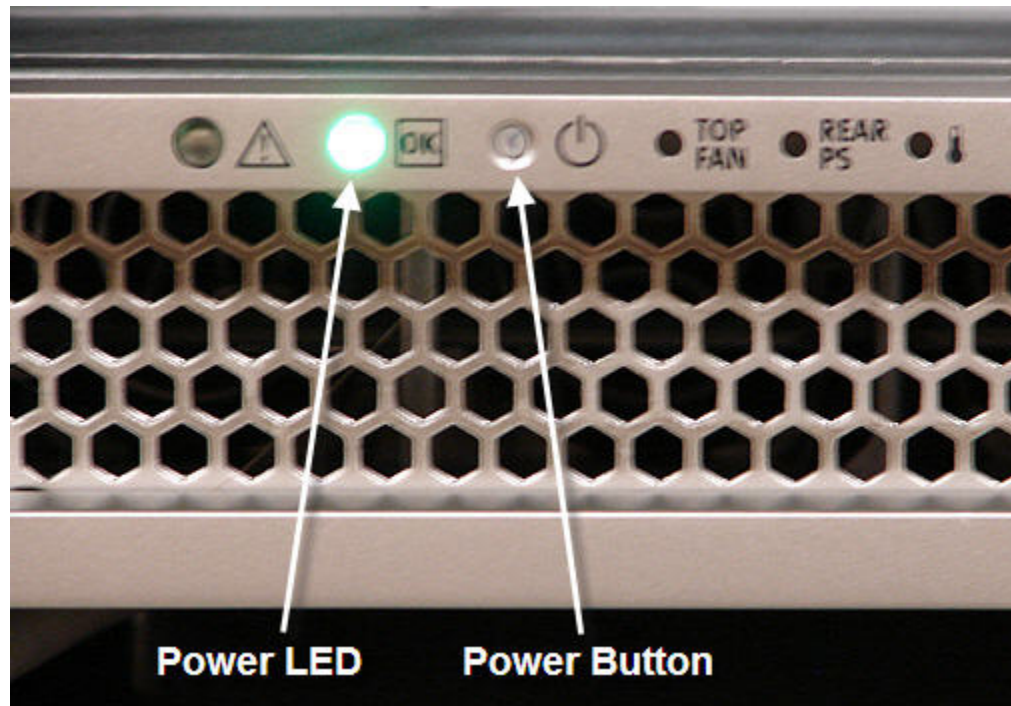


Illustration 2-8: ICN Power LED and Power Button (Sun 4170 and 4170M2)



2. If the OS freezes or is not responding, then the soft method did not work properly. In this case, hold the power button down for six seconds to force the power off.
3. For Dell ICN momentarily press the power button. The green LED is also the power button.

NOTICE

Holding the power button down for several seconds will bring it down hard.

Chapter 3 Upgrade Procedures

1 Sun ICN Upgrade Procedures

1.1 Overview

This procedure provides instructions on for removing existing Image Compute Nodes (ICNs) Sun 4100, Sun 4170, and Sun 4170 M2 to be replaced with the Dell R-630.

1.2 Removal Procedures

1. Make sure that the power to the ICN(s) have been turned off. Refer to Chapter 1.
2. Perform LOTO on the reconstruction equipment. See Chapter 3, Section 2.5 of the *MR Service Safety Manual*, PN 5452735.

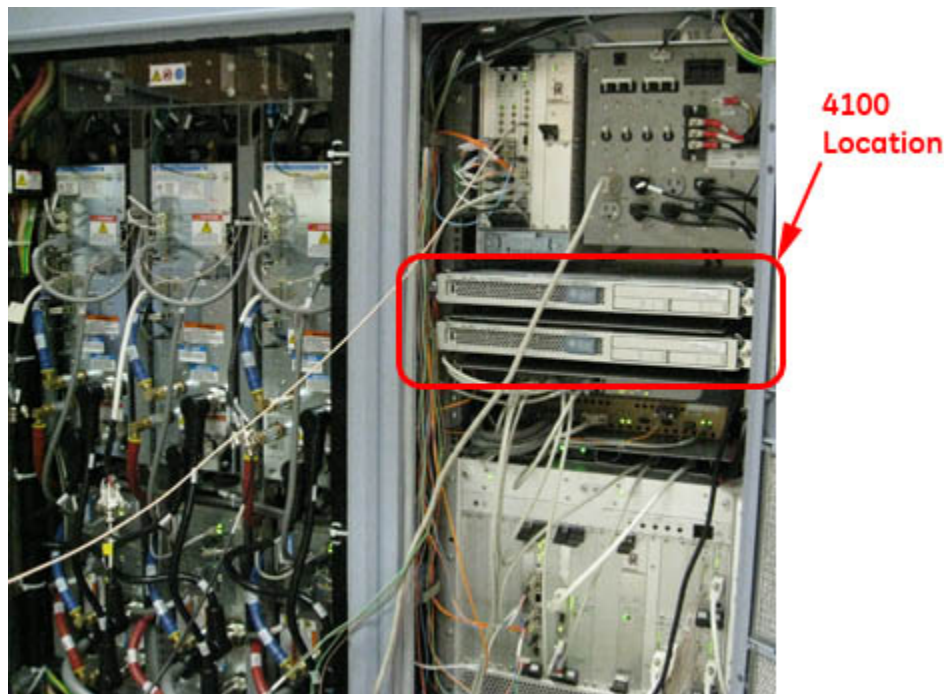
Refer to one of the following two procedures for removal of the existing ICN..

- [Section 1.2.1](#) for removal of Sun 4100 ICN
- [Section 1.2.2](#) for removal of Sun 4170 ICN

1.2.1 Removing ICN Sun 4100

Location of ICNs in PGR cabinet.

Illustration 3-1: ICN Location - 4100



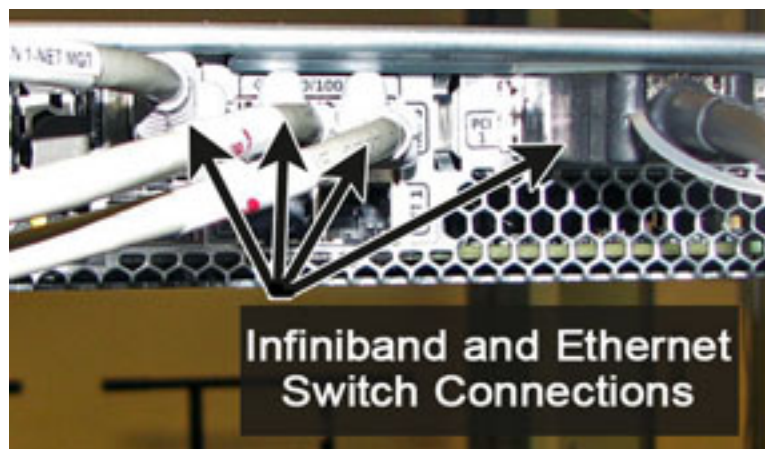
1. On the front of the ICN, undo the thumbscrew on each side of the ICNs being removed.

Illustration 3-2: Thumbscrews on ICN Front - 4100



2. Slide out the ICN until the stop is reached at the end of the side rails.
3. Remove all the connections at the back of the ICN.

Illustration 3-3: Cable Connections on Sun 4100 ICN

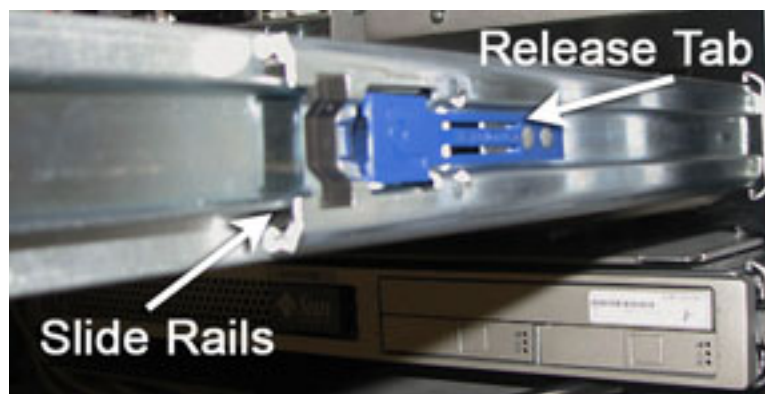


CAUTION

Possible personal injury!
The Sun 4100 ICN weighs approximately 40 pounds.
Two people are required to lift it.

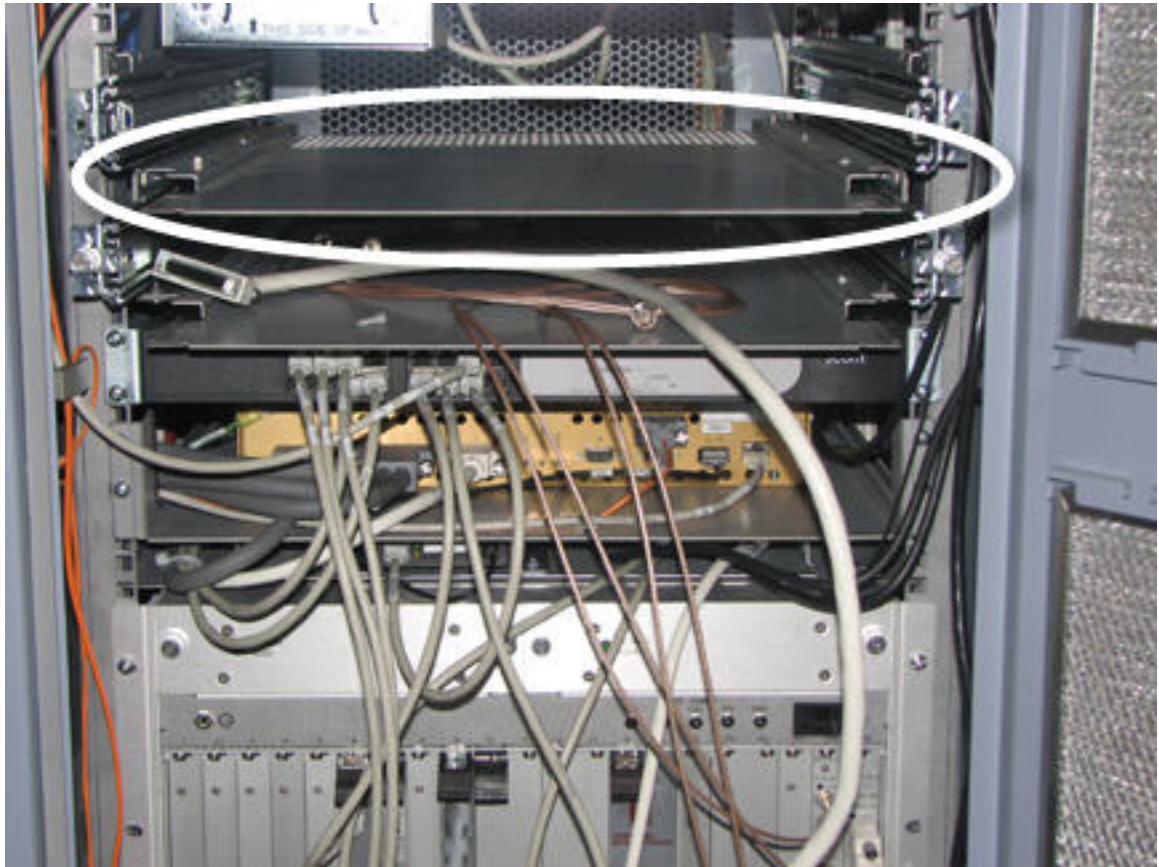
4. Locate the two release tabs on the slide rails. Press both release tabs toward the center of the ICN while pulling the ICN forward. Pull the ICN forward until the rails release and the ICN is free. Place the ICN to the side in a safe area.

Illustration 3-4: Slide Rails and Release Tab



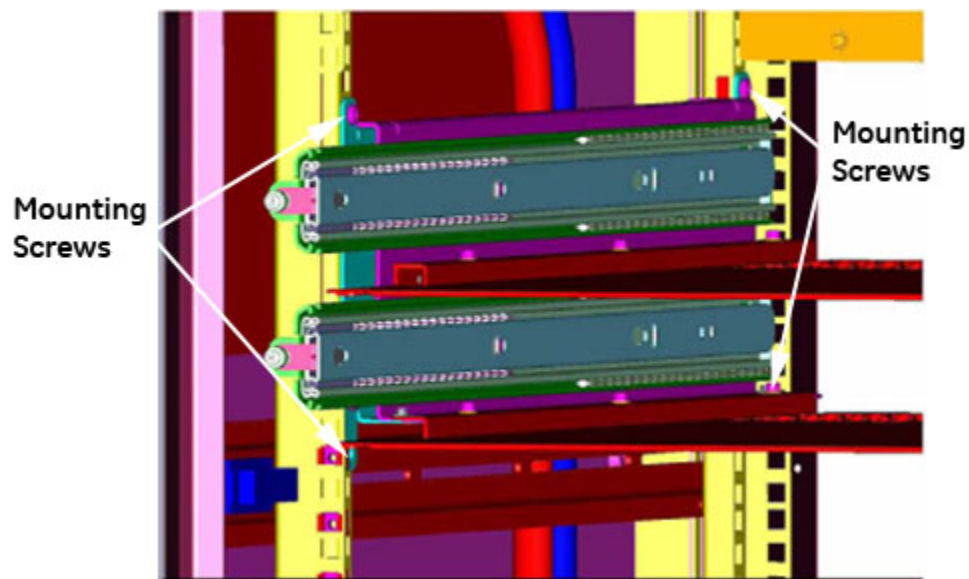
5. Remove the ICN trays from the ICN mounting brackets.

Illustration 3-5: Sun 4100 ICN Trays



6. Unfasten the screws securing the ICN mounting bracket/slide assemblies from the vertical rails and remove the assemblies from each side of the cabinet.

Illustration 3-6: Side Mounting Brackets for 4100

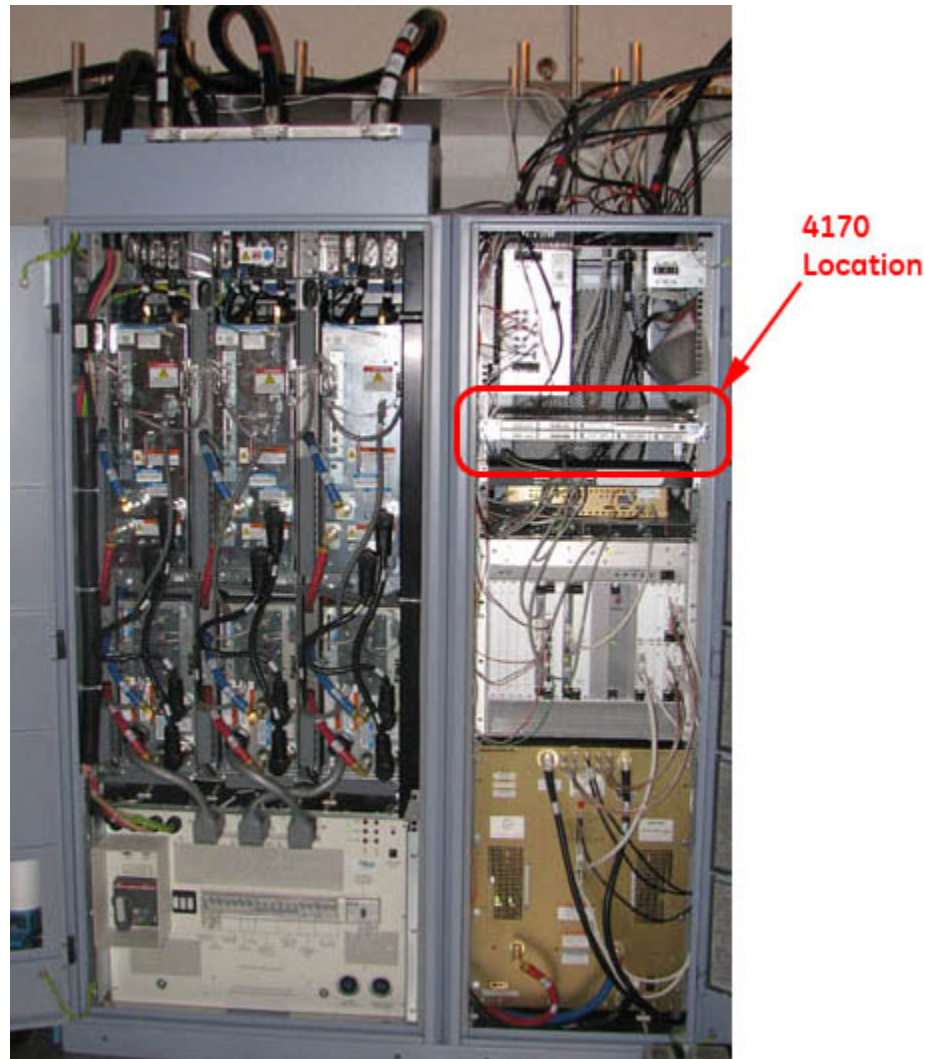


7. Remove and save all clip nuts that were used to fasten the above slides and bracket assemblies. They will be reused for installing the new ICN.

1.2.2 Removing ICN 4170

Location of ICN in PGR cabinet.

Illustration 3-7: ICN Location - 4170



1. On the front of the ICN, undo the thumbscrew on each side of the ICN to be removed.

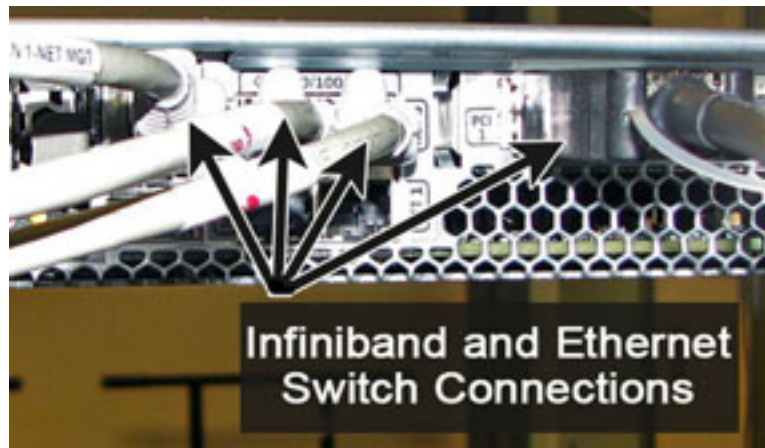
Illustration 3-8: Thumbscrews on ICN Front - 4170



2. Slide out the ICN until the stop is reached at the end of the side rails.

3. Remove all the connections at the back of the ICN.

Illustration 3-9: Cable Connections on Sun 4170 ICN



CAUTION

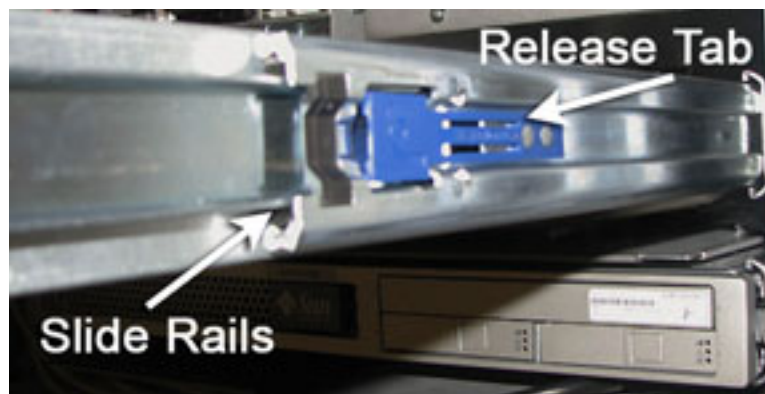
Possible personal injury!

The Sun 4170 ICN weighs approximately 40 pounds.

Two people are required to lift it.

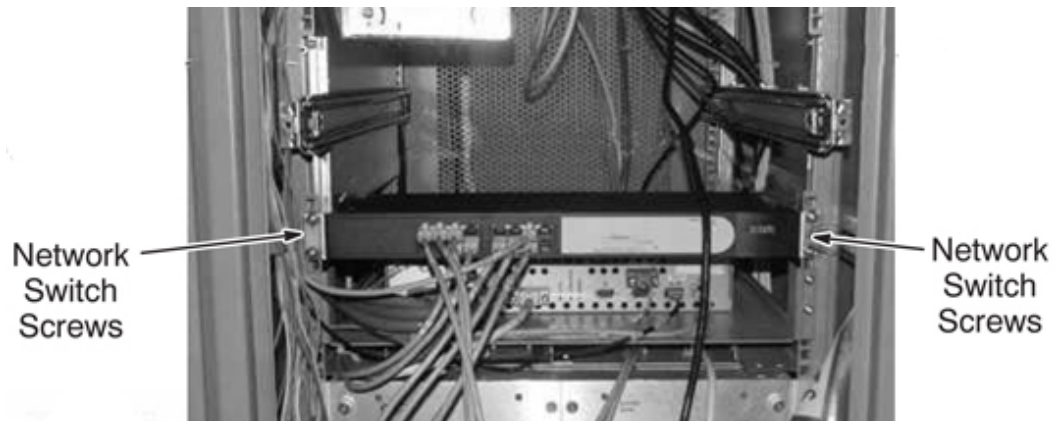
4. Locate the two release tabs on the slide rails. Press both release tabs toward the center of the ICN while pulling the ICN forward. Pull the ICN forward until the rails release and the ICN is free. Remove the ICN and place to the side in a safe area.

Illustration 3-10: Slide Rails and Release Tab



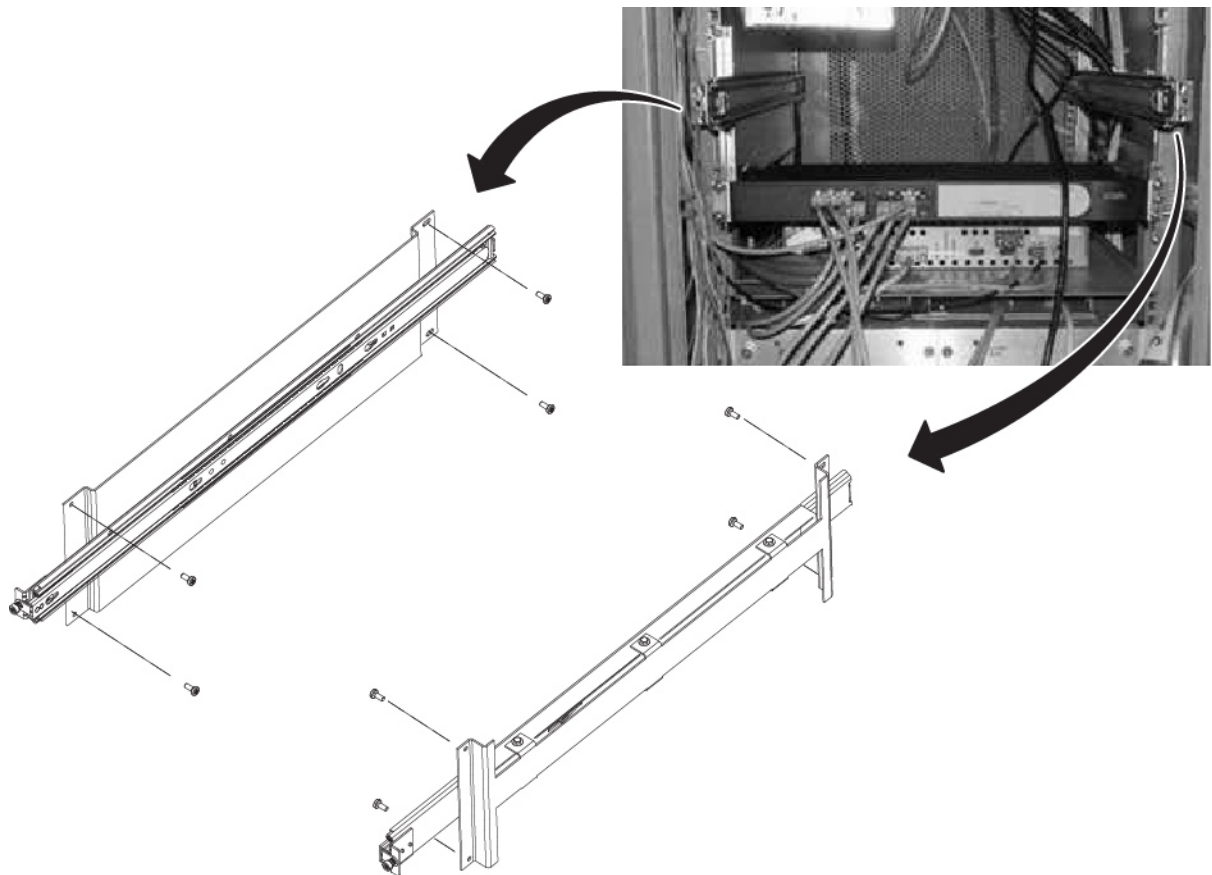
5. Loosen the mounting screws for the network switch. Slide the switch down to give access to the mounting location for the slide rails for the 4170 ICN.

Illustration 3-11: Network Switch Screws



6. Unfasten screws and remove slide and bracket assemblies from each side of cabinet.

Illustration 3-12: ICN Slide and Bracket Removal

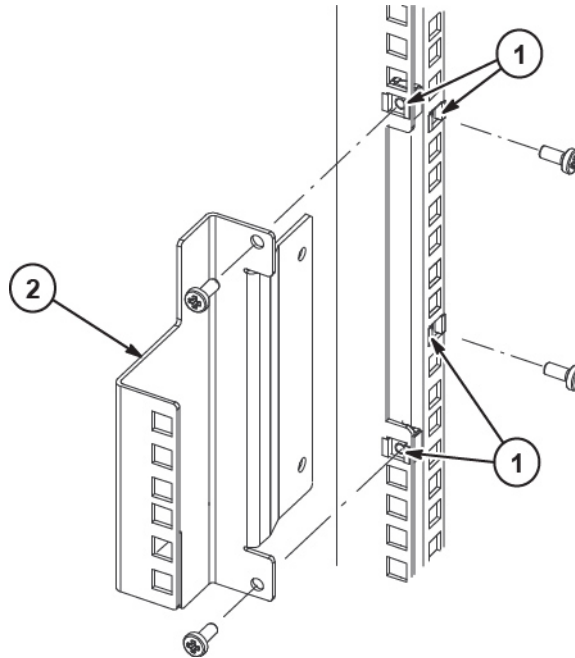


7. Remove and save all clip nuts that were used to fasten the above slides and bracket assemblies. They will be reused for installing the new ICN.

1.3 Install Dell R-630 ICN and Mounting Hardware

The installation of the Dell R-630 will require attaching new mounting brackets (5496069) on the front vertical cabinet rails.

Illustration 3-13: ICN Bracket Installation



1. Install the clip nuts to the locations indicated around the cut out on the front vertical rack rail as shown in the above illustration. Left side vertical rack rail is shown. Repeat for right side rail.
2. Attach brackets to each rail using the previously removed screws.

Illustration 3-14: Ethernet Switch



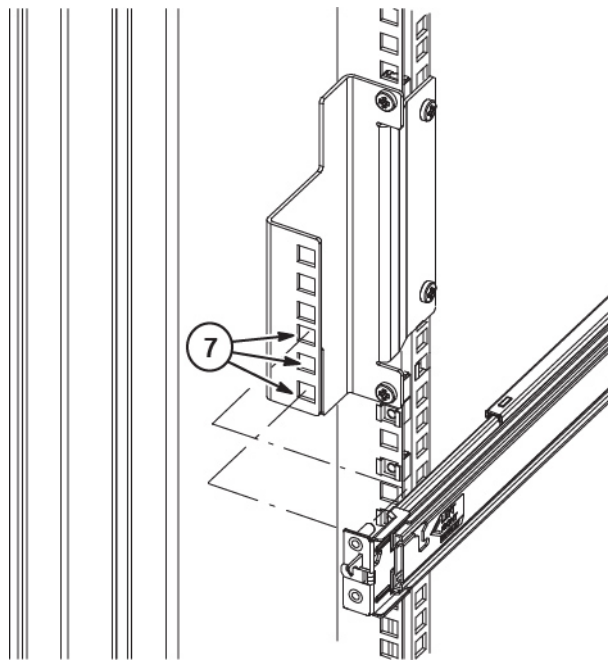
3. Reattach the Ethernet Switch.
4. New cables will be installed later in procedure.

Illustration 3-15: ICN Slide Rails



5. Locate the Dell mounting rails.
6. Remove the orange shipping clips.

Illustration 3-16: Align Front of Slide

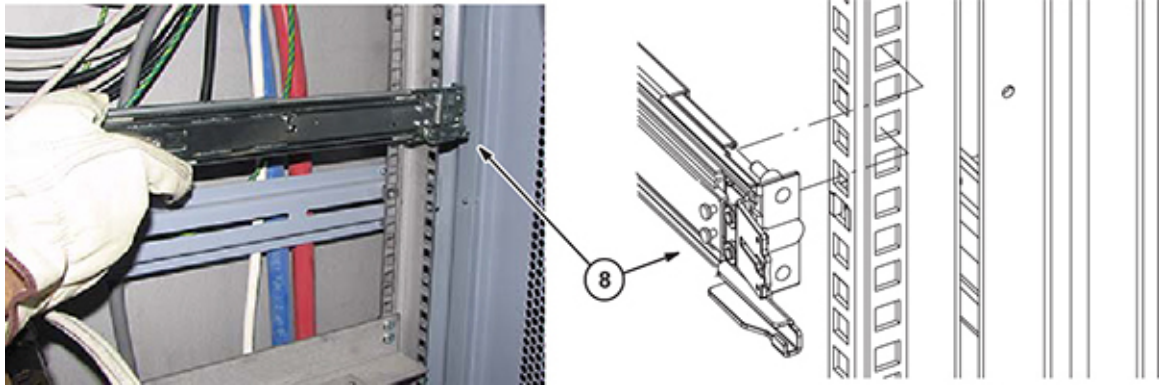


7. Position the left slide rail labeled FRONT facing inward and align the end of the rail to seat in the bottom three holes of the installed bracket.

NOTICE

Verify that the new back rail bracket is aligned to the proper holes prior to seating. Use a level to make sure that the rail is aligned to the correct holes. It is difficult to remove the back rail bracket after it has been installed.

Illustration 3-17: Secure Rear of Slide



8. At the back end of the rail, align and engage the slide rail by pulling it outward, until it fully seats into the rear vertical cabinet rack rail and the latch clicks into place

Illustration 3-18: Secure Front of Slide



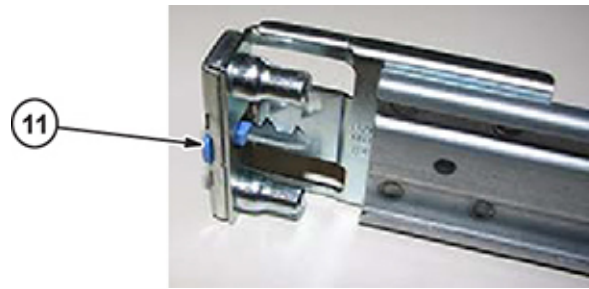
9. Engage the front end of the slide rail into the vertical cabinet rack rail.

Illustration 3-19: Install Right Side Slide Rail



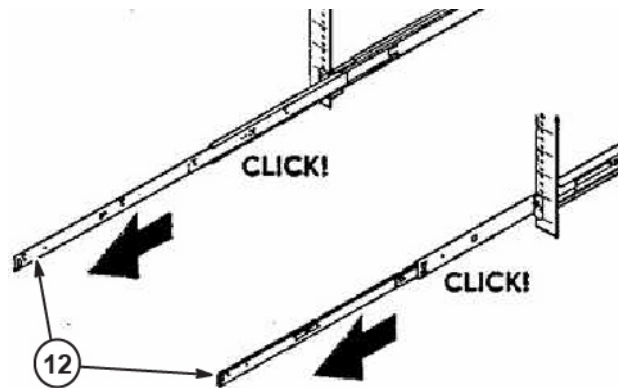
10. Repeat previous steps for installing right side slide rail.

Illustration 3-20: Release Tab



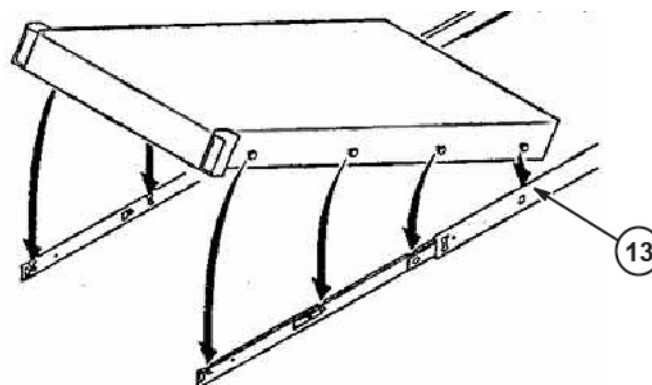
11. The blue tab on the end of the slides must be flat with the end surface of the rail for the slide rail to be completely connected to the vertical cabinet rail or bracket. If rails need to be removed, pull the blue latch release tab on the end of each rail and unseat. The blue tab mechanism can easily be observed on the front of the rails.

Illustration 3-21: Slide Out Rails



12. Pull the inner slide rails out of the rack until they lock into place.

Illustration 3-22: Move ICN to Rails

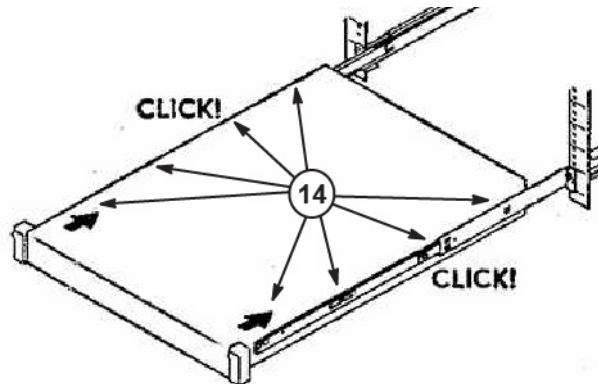


CAUTION

Possible personal injury!
The ICN weighs approximately 40 pounds.
Two people are required to lift it.

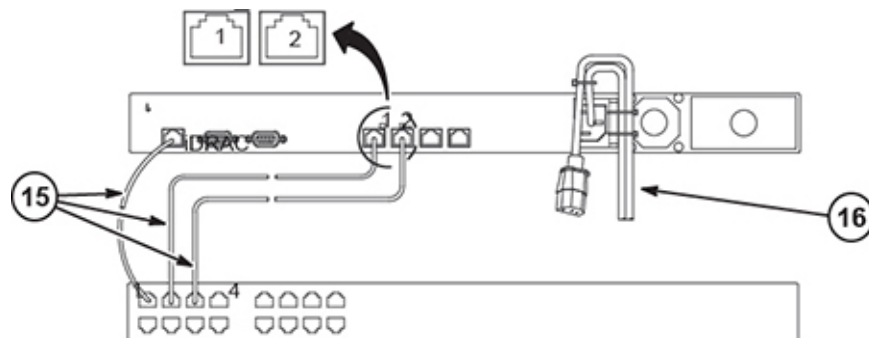
13. Remove the new ICN from packaging. Position the rear rail standoff on each side of the ICN and lower into the rear J-slots on the slides.

Illustration 3-23: Secure ICN in Rails



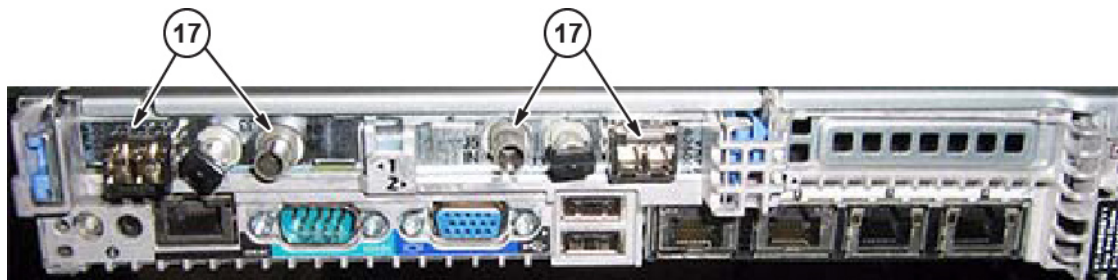
14. Rotate the ICN module downward until all eight rail standoffs are seated in the J-slots. Push the ICN module inward until the two lock levers, one on each side, click into place.

Illustration 3-24: Install ICN Cables



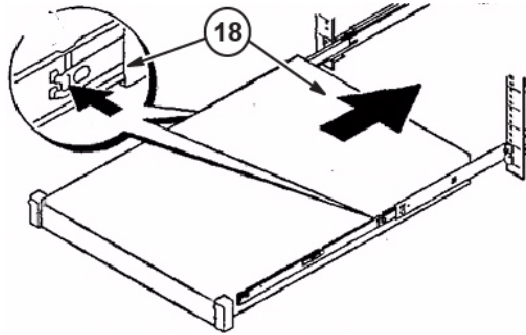
15. Remove the existing 3 Ethernet cables that run between the ICN and Ethernet switch and install new ones that are provided.
16. Connect power cables to the power port(s) on the ICN.

Illustration 3-25: Connect Fiber Optic Cables



17. Reconnect the existing fiber optic cables to J1 and J5 as marked. Two sets of fiber optic cables will be connected if the system has a 32 Channel ICN as shown in the illustration.

Illustration 3-26: Move ICN Into Cabinet



18. Press the slide-release lock buttons on both rails and carefully slide the module into the cabinet.

2 Dell ICN Upgrade Procedures

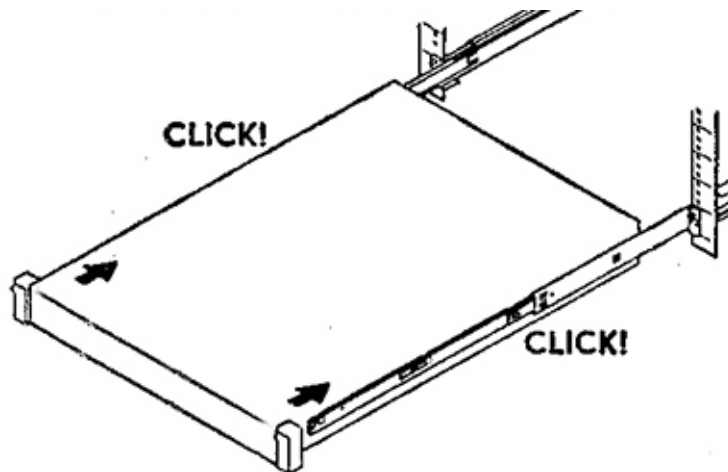
2.1 Overview

This procedure provides instructions on for removing existing Dell R-620 or Dell R-630 32 Channel Image Compute Node (ICN) and replaced with the Dell R-630 32 Channel Plus ICN.

2.2 Removal Procedures

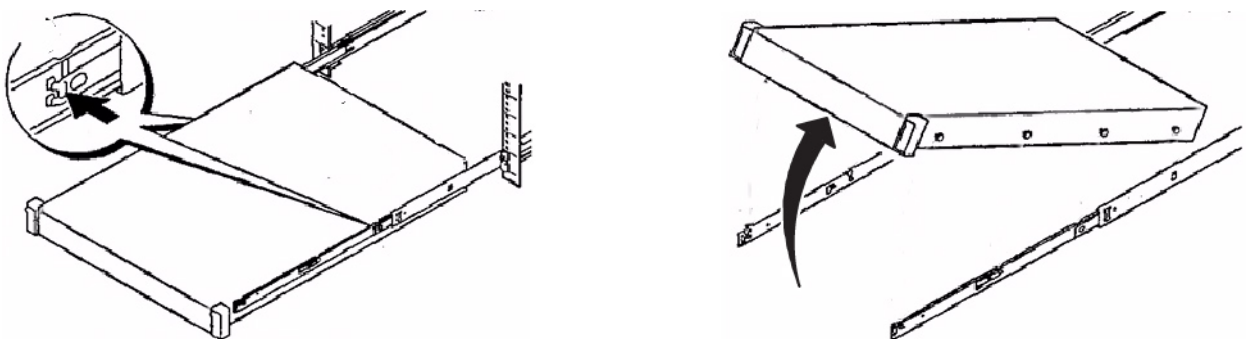
1. Make sure that the power to the ICN has been turned off. Refer to Chapter 1.
2. Perform LOTO on the reconstruction equipment. See the *MR Service Safety Manual*, PN 5452735.

Illustration 3-27: Move ICN out of Cabinet



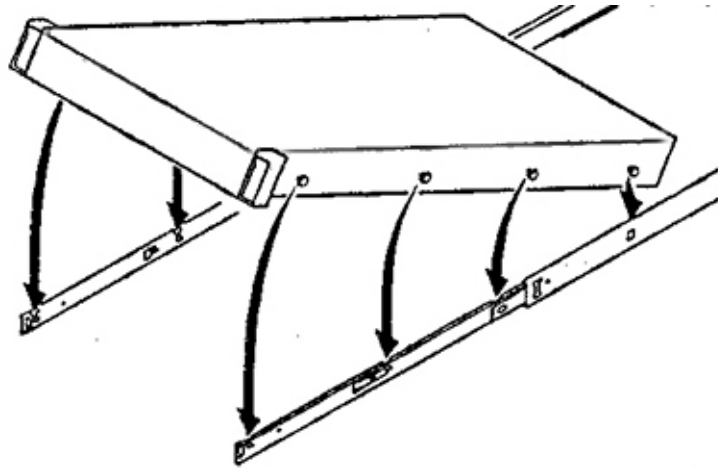
3. Pull the ICN module slide rails out of the rack until they lock into place.
4. Disconnect all cables from the rear of the ICN module.

Illustration 3-28: Remove Dell ICN



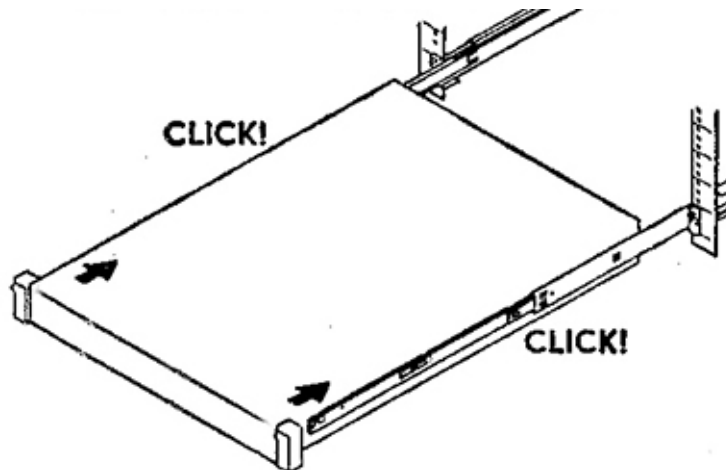
5. Locate the two release tabs on the slide rails. Press both release tabs toward the center of the ICN while pulling the ICN forward. Pull the ICN forward until the rails release and the ICN is free. Pivot ICN upwards and remove.

Illustration 3-29: Move New Dell Plus ICN to Rails



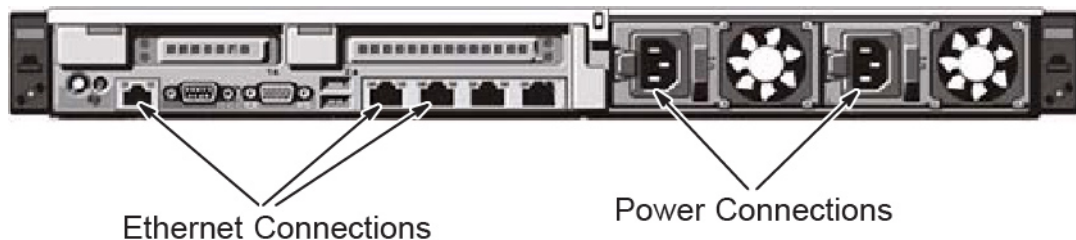
6. Remove the new Dell 32 Channel Plus ICN from packaging. Position the rear rail standoff on each side of the ICN and lower into the rear J-slots on the slides.

Illustration 3-30: Secure ICN in Rails



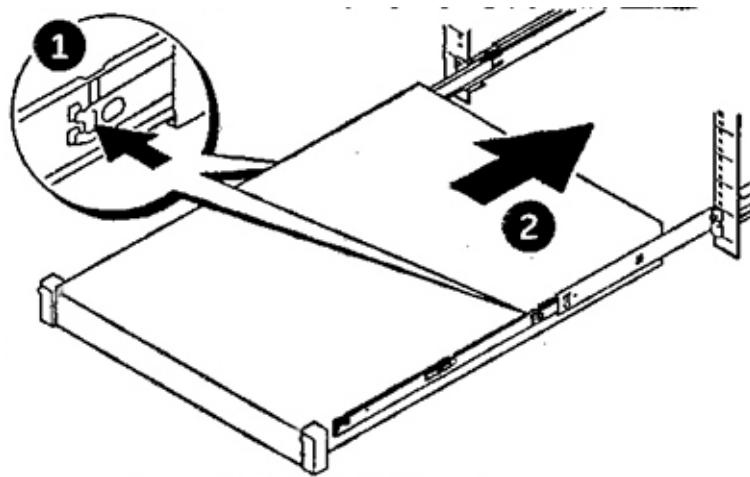
7. Rotate the ICN module downward until all eight rail standoffs are seated in the J-slots. Push the ICN module inward until the two lock levers, one on each side, click into place.

Illustration 3-31: Connect cables to ICN



8. Reconnect the 3 ethernet cables. Connect the two power cables.

Illustration 3-32: Move ICN into Cabinet



9. Press the slide-release lock buttons on both rails and carefully slide the module into the cabinet.

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Chapter 4 Power On

1 Power On Procedure

1. Remove LOTO from the reconstruction equipment. See the *MR Service Safety Manual*, PN 5452735.

NOTE: The Dell ICN (GEN5) is only compatible with DV25 software. The Dell ICN (GEN6) is only compatible with DV25.1_R01 software.

2. On the Common Service Desktop, go to Utilities > Toolbox > VRE > Power Control. Click [POWER ON ICN] for the selected ICN. The new ICN is automatically powered on.

NOTE: The Dell ICN will display the boot status on the LED readout, which is located on the front panel of the ICN.

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Chapter 5 ICN Reconfiguration

1 ICN Reconfiguration

The Dell ICN (GEN6) is only compatible with DV25 software.

1. Gen 6 ICNs are supported only on DV25.1_R01 and later s/w. If needed perform a DV25 Host Software Load.
2. If already on DV25.1_R01 or later s/w, Perform ICN reconfiguration per Service procedures.
3. After ICN Reconfiguration completes, reboot the system.
4. Perform a Check Scan and ensure system scans successfully.
5. Perform a SAVEINFO to save new updated vre.cfg files onto Save Info Media.

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